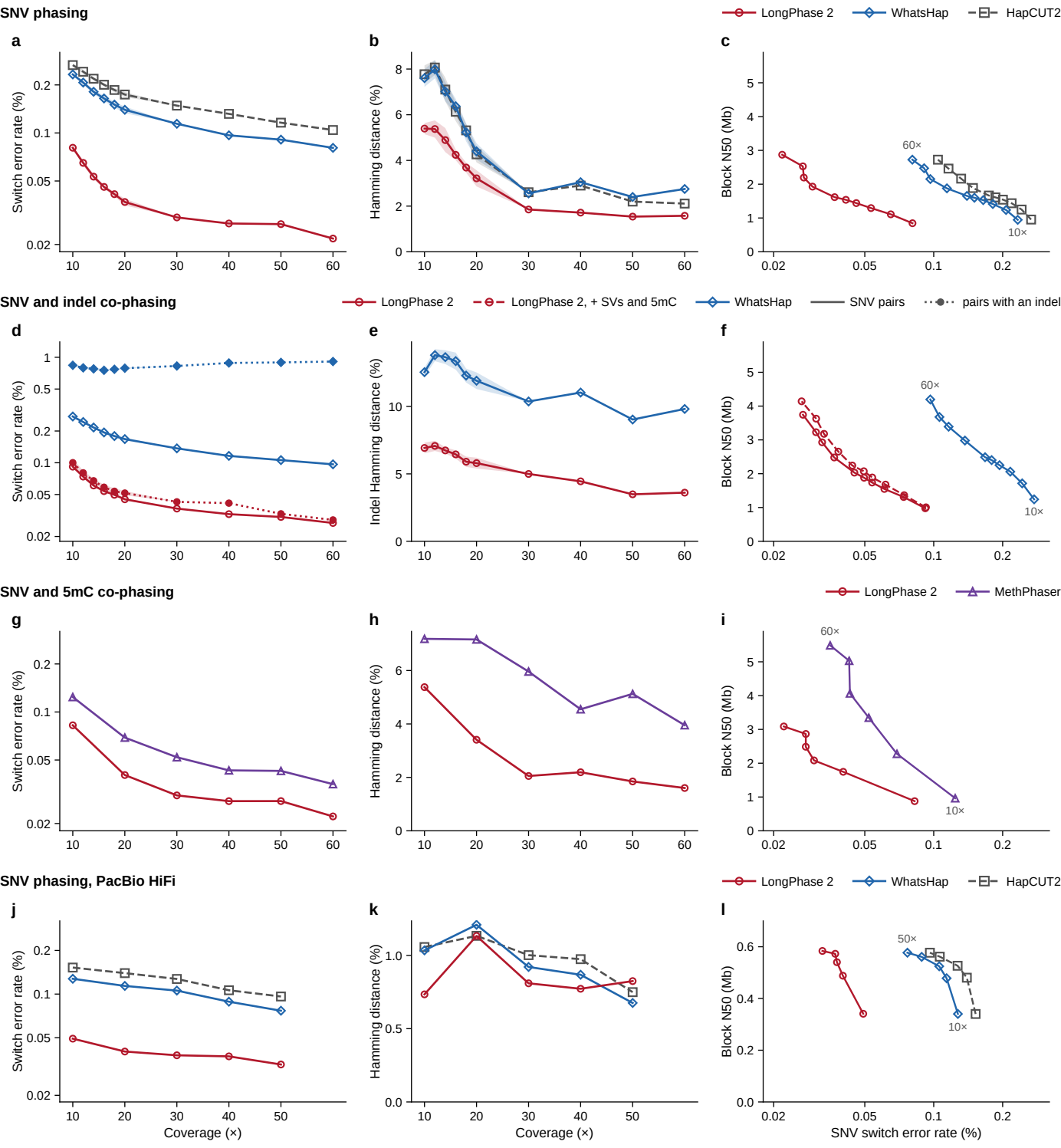
SNV phasing
LongPhase 2
WhatsHap
HapCUT2
a
b
c
Switch error rate (%)
Hamming distance (%)
Block N50 (Mb)
60×
10×
SNV and indel co-phasing
LongPhase 2
LongPhase 2, + SVs and 5mC
WhatsHap
SNV pairs
pairs with an indel
d
e
f
Switch error rate (%)
Indel Hamming distance (%)
Block N50 (Mb)
60×
10×
SNV and 5mC co-phasing
LongPhase 2
MethPhaser
g
h
i
Switch error rate (%)
Hamming distance (%)
Block N50 (Mb)
60×
10×
SNV phasing, PacBio HiFi
LongPhase 2
WhatsHap
HapCUT2
j
k
l
Switch error rate (%)
Hamming distance (%)
Block N50 (Mb)
50×
10×
Coverage (×)
Coverage (×)
SNV switch error rate (%)